# Philippine Burdeau

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## Summary

Stanford PhD (graduating Summer 2026) combining quantitative training in applied mathematics (École Polytechnique, France) with technical expertise in climate tech and energy systems. Authentic, collaborative and people-oriented, with experience in strategy consulting and techno-economic analysis. Seeking roles bridging technical depth and strategic decision-making in deep tech, climate, or energy (e.g., technical due diligence, product or technical program management, or research operations).

## Experience

**Stanford University – Research Assistant** — Sept 2022 – Present
*Environmental Assessment and Optimization Group* — Stanford, CA

- Authored or co-authored 8 papers (3 first-author, 6 published, 2 in preparation). First paper as lead author published in *Nature Communications* (2025) and won Best Poster Award at Stanford; now independently leading 2 additional research projects from conception to submission.
- Built end-to-end ML and statistical pipelines: trained models on >2M oil & gas production records, developed MLE and Bayesian hierarchical inference frameworks, and fused heterogeneous satellite, aerial, and ground-based sensor datasets for methane emissions quantification.
- Supported controlled release field operations in Arizona (2022), then took on expanded responsibilities for a campaign in France (2024): designed the release schedule and coordinated on-site operations across 15+ industry and academic partners.
- Mentored 2 junior PhD students and supervised student project teams as TA for a PhD-level optimization course.

**Insight M – Research Intern** — Jun – Aug 2024
*Methane emissions management startup* — Sunnyvale, CA

- Enhanced stochastic emissions model and developed it into production-ready software.

**Kearney – Business Analyst Intern** — May – Nov 2021
*Global management consulting firm* — Paris, France

- Authored a techno-economic report on carbon emissions management for the Energy Transition Institute, synthesizing data from 50+ sources on monitoring, reporting, and verification technologies.
- Conducted due diligence for an automotive insurer evaluating startup partnerships and M&A opportunities; developed market sizing, competitive landscape analysis, and strategic recommendations for the client's executive team.

**France Stratégie – Research Intern** — Apr – Aug 2020
*Policy advisory agency to the French Prime Minister* — Paris, France

- Modeled carbon abatement costs across sectors, contributing to France's national carbon valuation framework following the Quinet Report on the value of climate action.

## Education

**Stanford University, Doerr School of Sustainability** — 2022 – 2026
PhD in Energy Science and Engineering (Prof. Adam Brandt) — Stanford, CA

- Dissertation: Statistical methods for quantifying methane emissions from measurement data
- Relevant coursework: Convex Optimization, Applied Statistics, Decision Making Under Uncertainty, and graduate-level energy systems and computational methods courses.

**Università Bocconi** — 2020 – 2022
MSc in Politics and Policy Analysis — Milan, Italy

**École Polytechnique** – **Ingénieur Polytechnicien Program** — 2017 – 2021
#1 French school of engineering; MSc in Applied Mathematics and Physics — Palaiseau, France

- Coursework incl. Advanced Probability, Statistical Physics, Quantum Mechanics

**Lycée Sainte-Geneviève (MPSI – MP*)** — 2015 – 2017
#1 French preparatory school; two-year undergraduate intensive program for the *Grandes Écoles* — Versailles, France

## Selected Publications – full list on Google Scholar

**Burdeau P.** et al. "High-resolution national mapping of natural gas composition substantially updates methane leakage impacts." *Nature Communications*, 2025. `link`

**Burdeau P.** et al. "Statistical inference of intermittent methane emissions from heterogeneous measurements." In preparation for *J. R. Stat. Soc. A*, 2026.

**Burdeau P.** et al. "Bayesian hierarchical multi-scale fusion for methane emissions estimation." In preparation, 2026.

**4 co-authored papers on controlled release experiments** Sherwin et al., *Atmos. Meas. Tech.*, 2024 (`link`); El Abbadi et al., *ES&T*, 2024 (`link`); Chen et al., *ACS ES&T Air*, 2024 (`link`); McManemin et al., *Atmos. Meas. Tech.*, 2026 (`link`).

## Honors

**Best Poster Award**, Stanford ESE in 2050 (1st / 33) 2025

**Fulbright Scholar** (Fondation Monahan) 2022

**Best Scientific Group Project Prize**, École Polytechnique (awarded to 3 groups out of 120) 2019

## Service and Leadership

**Teaching**: TA, PhD-level Optimization of Energy Systems | Guest Lecturer, methane emissions course 2024–25

**Journal reviewer**: *ACS ES&T Air*, Tackling Climate Change with ML (ICLR), CCAI Innovation Grants 2024–25

**Seminar organizer**: Methane Technology Alliance seminar series (Stanford & LBNL) 2022–Present

**French Navy Officer**: human and military training on a naval base in New Caledonia 2017–18

## Skills and Interests

**Languages:** English (TOEFL 110), French (native), Italian
**Technical:** Python, R, SQL, Stata, Matlab, Excel/VBA

**Extracurriculars:** Piano, running, writing
Editorial Committee, École Polytechnique alumni journal